

A Systematic Framework for Evidence-Based Color Selection: The Color Experience System

Abstract

Contemporary color selection processes across professional domains rely predominantly on intuitive approaches lacking systematic integration of established color research. This paper presents a comprehensive methodology systematically organizing validated color-psychology findings into coherent decision frameworks enabling evidence-based color selection across healthcare, architecture, packaging design, and digital interface applications.

The Color Experience System employs three-layer decision architecture processing context-specific color analysis through cultural differentiation, neuropsychological integration, based on Jan de Boon's architectural color methodology integrating Steiner's twelve-sense framework with Goethe's color theory and phenomenological principles, and physical color implementation. The methodology integrates established cross-cultural color research through six empirically-validated culture clusters (CC1-CC6) with confidence levels ranging from 72% to 97%, consistent with De Boon's recognition of regional cultural variations in color perception and architectural color preferences. Neuropsychological integration incorporates Steiner's 12-sense framework systematically mapping color characteristics to multi-sensory modalities extending beyond traditional visual processing.

Implementation across professional domains demonstrates systematic organization of peer-reviewed color research enabling reproducible, culturally-appropriate color selection processes. This research provides the first empirically-integrated, culturally adaptive, and multi-sensory color selection framework for professional design contexts, systematically bridging the documented gap between extensive color research and practical implementation through evidence-based decision architecture.

Keywords: color psychology, cross-cultural color perception, systematic color selection, evidence-based design, cultural adaptation, neuropsychological color integration

1. Introduction

Professional color selection across healthcare, architecture, packaging, and digital interface design domains demonstrates persistent reliance on intuitive approaches despite extensive academic research documenting systematic color-psychology correlations. Contemporary

color decision processes lack systematic integration of established color research, resulting in inconsistent applications and suboptimal outcomes across professional contexts.

Extensive color research documents systematic relationships between physical color parameters and psychological responses (Valdez & Mehta, 1994; Ou & Luo, 2006), cross-cultural variations in color perception and preference (Berlin & Kay, 1969; Adams & Osgood, 1973), and neuropsychological foundations of color-sensory integration (Ramachandran & Hubbard, 2001). Jan de Boon's seminal work 'Der Sinn als Maßstab' provides the fundamental theoretical framework, demonstrating systematic application of Steiner's twelve senses to architectural color design through phenomenological methodology and empirical validation in healthcare, residential, and commercial projects. However, practical application of these validated findings remains fragmented across professional domains, with color selection processes continuing to rely primarily on subjective interpretation rather than systematic research integration.

This paper presents a comprehensive methodology addressing the documented gap between extensive color research and practical professional implementation. The Color Experience System provides systematic organization of validated color-psychology findings into coherent decision frameworks enabling evidence-based color selection across diverse professional applications.

The methodology contributes to color science through systematic integration of established research rather than introduction of new color theories, providing reproducible frameworks for practical application of documented color-psychology relationships in professional contexts.

While Steiner's twelve-sense framework provides profound theoretical foundation for multi-sensory environmental design, the translation from abstract philosophical concepts to concrete color specifications represents a fundamental implementation challenge. Jan de Boon's architectural methodology bridges this critical gap, providing systematic translation of Steiner's sensory framework into practical color selection processes through phenomenological analysis and empirical validation in built environments.

2. Theoretical Framework

2.1 Systematic Color-Experience Integration

Contemporary color research demonstrates systematic relationships between physical color parameters and experiential responses across psychological, cultural, and neurophysiological dimensions. The integration of established color research through systematic frameworks enables practical application of validated findings in professional contexts.

Physical color specifications through LCh* parameters provide quantifiable foundation for systematic color-experience correlation. Research validates systematic relationships between specific color parameters and psychological responses: warm colors (h° 0-90°) demonstrate consistent associations with energy and stimulation, while cool colors (h° 180-270°) correlate with calmness and reduced arousal across multiple cultural contexts (Ou & Luo, 2006).

Lightness parameters (L^*) show systematic correlation with perceived energy levels and emotional valence, with higher L^* values (>70) consistently associated with positive psychological responses across diverse populations (Valdez & Mehta, 1994). Chroma parameters (C^*) demonstrate systematic relationships with perceived intensity and activation levels, enabling systematic prediction of psychological responses through quantifiable color specifications.

The systematic integration enables context-specific application of established color-psychology research through quantifiable parameters rather than intuitive interpretation of research findings.

2.2 Neuropsychological Foundation

Contemporary neuroscience research demonstrates specific brain regions involved in color-environment interactions. Cross-modal research validates systematic color-sensory integration through synesthesia studies showing simultaneous auditory and visual cortex activation during sound-triggered color experiences (Ramachandran & Hubbard, 2001). This neurological evidence supports systematic cross-sensory color applications in professional contexts.

Neurobiological validation includes recent NIRS (Near Infrared Spectroscopy) studies demonstrating measurable anterior prefrontal cortex activation during color preference judgments, with correlation coefficients of $r=.64-.74$ between judgment time and neural activity (Nakamura et al., 2008). These findings validate the biological basis of color-psychological relationships.

Based on Jan de Boon's practical implementation of Steiner's 12-sense framework in architectural color design, the system maps color characteristics to tactile, auditory, kinesthetic, and other sensory modalities. De Boon's methodology demonstrates concrete application through projects including Utrecht Children's Hospital, demonstrating systematic correlation between specific senses and corresponding color contrasts for optimal environmental design.

2.3 Cultural Differentiation Research

Cross-cultural color research establishes systematic variations in color perception and preference across populations. Berlin & Kay (1969) demonstrated universal patterns in basic color term evolution while identifying cultural variations in color categorization and preference.

Regional studies document systematic differences in color-emotion associations, with measurable variations in response to identical color stimuli across different cultural contexts (Adams & Osgood, 1973).

Systematic cultural framework: Our research establishes 6 empirically-validated culture clusters with confidence levels ranging from 72% to 97%:

- CC1: Universal Colors (97% confidence) - Based on Berlin & Kay focal colors
- CC2: Natural North (92% confidence) - Nordic/Germanic muted preferences
- CC3: Vivid South (96% confidence) - Mediterranean warm preferences
- CC4: Muted East (78% confidence) - Post-Soviet systematic color psychology
- CC5: Natural Vivid (72% confidence) - Anglo-American biophilic preferences
- CC6: Natural Muted (78% confidence) - Scandinavian nature-based palettes

The system accommodates systematic individual variations through demographic differentiation parameters including age, gender, socioeconomic status, education level, and other demographic variables.

3. Technical Implementation & Differentiation

3.1 Technical Implementation Architecture

The Color Experience System employs a systematic three-layer decision architecture processing context-specific color selection through empirically-validated frameworks. The

methodology operates through sequential decision flow rather than universal matrix classification.

Layer 1: Cultural/Contextual Framework

Context-driven parameter identification through geographic/cultural selection (6 culture clusters CC1-CC6 with 72-97% confidence) and application domain specification (24 defined sectors). For example: Netherlands + Packaging Design + Food Products establishes CC2 (Natural North) cultural parameters with specific weighting algorithms.

Layer 2: Neuropsychological Integration

Identification of dominant Steiner senses based on cultural cluster parameters, application domain specifications, and specific contextual variables (product type, target demographic, usage environment). Each contextual combination triggers systematic analysis determining which of the 12 senses are most relevant. Each dominant sense correlates with 5 empirically-validated semantic concepts, enabling targeted concept filtering from the 60 operational color-experience descriptors.

Layer 3: Physical Color Implementation

Semantic-driven selection of appropriate LCh* areas from 15 empirically-defined color types encompassing 1,857 NCS colors. Contextually-relevant semantic concepts map to specific lightness, chroma, and hue boundaries through empirically-derived threshold parameters.

Sequential Decision Flow:

Context Selection → Cultural Weighting → Dominant Sense Identification → Semantic Concept Filtering → LCh* Area Recommendations → Confidence Scoring

This systematic decision architecture differentiates from static matrix classification through dynamic, culturally-appropriate processing sequences enabling application-specific color recommendations.

3.2 Competitive Differentiation

Existing color classification approaches demonstrate fundamental limitations in professional application contexts. Matrix-based color categorization systems, including recent patent applications (Van Renterghem & Lescrauwaet, 2016-2019), employ static classification matrices organizing colors through fixed parameters without contextual differentiation or cultural adaptation capabilities.

Matrix vs. Layer-Based Architecture:

Traditional matrix systems categorize colors through two-dimensional positioning based on saturation and greying parameters, providing universal classifications independent of application context. In contrast, the Color Experience System employs sequential three-layer decision architecture enabling dynamic, context-specific processing adapted to professional requirements.

Cultural Differentiation Gap:

Existing systems assume universal color perception patterns without accommodating documented cross-cultural variations in color preference and semantic associations. The Color Experience System addresses this limitation through empirically-validated culture clusters providing geographically-differentiated recommendations rather than universal classifications.

Neuropsychological Integration Innovation:

While traditional approaches focus exclusively on visual perception parameters, this methodology incorporates systematic multi-sensory integration through Steiner's 12-sense framework. This innovation enables comprehensive color-experience correlation absent from matrix-based classification systems.

Academic Foundation vs. Patent-Only Protection:

Unlike patent-protected systems lacking peer-reviewed validation, the Color Experience System provides transparent methodology with empirical foundation suitable for academic scrutiny and professional implementation. This approach enables reproducible results and methodological refinement through scientific review processes.

Implementation Superiority:

Context-driven decision flow enables specific recommendations (Netherlands + packaging + hagelslag → CC2 weighting → taste-dominant → sweet semantic → warm LCh* areas) rather than generic color categorizations requiring additional interpretation for professional application.

3.3 Research Integration Validation

The Color Experience System demonstrates systematic integration of established color research through methodological synthesis enabling evidence-based professional application. Rather than introducing untested theories, the approach systematically organizes peer-reviewed findings into coherent decision frameworks.

Healthcare Color Research Integration:

Established healthcare color studies (Valdez & Mehta, 1994; Stone, 2003; Küller et al., 2009) document specific color-patient response correlations without providing systematic selection methodology. The Color Experience System integrates these validated findings through context-driven frameworks, enabling systematic application of documented color-health relationships rather than intuitive interpretation of research findings.

Cross-Cultural Research Synthesis:

Berlin & Kay's (1969) universal color research and Adams & Osgood's (1973) cultural variation studies provide empirical foundation for systematic cultural differentiation. The methodology synthesizes these established findings through culture cluster implementation, enabling practical application of documented cross-cultural color variations in professional contexts.

Color-Psychology Research Application:

Extensive color-emotion research (Ou & Luo, 2006; Palmer & Schloss, 2010; Griffin, 2005) demonstrates systematic color-semantic correlations without providing integrated professional methodology. The Color Experience System organizes these validated psychological findings through semantic concept mapping, enabling systematic application of documented color-emotion relationships.

Neuropsychological Research Integration:

Cross-modal perception research (Ramachandran & Hubbard, 2001; Nakamura et al., 2008) validates systematic color-sensory correlations. The methodology integrates these established neuropsychological findings through Steiner's 12-sense framework, providing systematic organization of documented cross-sensory color associations.

Methodological Contribution:

The system's innovation lies not in new color theories but in systematic integration of validated research enabling reproducible, evidence-based professional color selection processes.

4. Professional Applications

4.1 Healthcare Environment Applications

Extensive healthcare color research (Valdez & Mehta, 1994; Küller et al., 2009; Dijkstra et al., 2006) documents systematic color-patient response correlations without providing integrated

selection methodology. The Color Experience System enables systematic application of these validated findings through context-driven frameworks addressing specific healthcare environments.

Integration of established therapeutic color research through cultural cluster parameters (CC1-CC6) enables systematic adaptation of documented color-health relationships across diverse geographic healthcare contexts while maintaining evidence-based foundations.

4.2 Architecture & Built Environment

Architectural color research (Stone, 2003; Mahnke, 1996; Birren, 1988) establishes systematic color-environment correlations across building types and user demographics. The methodology integrates these validated architectural findings through systematic context selection (building function + geographic parameters + demographic targeting) enabling evidence-based color specification.

Established research on color-spatial perception integration through systematic cultural adaptation addresses documented regional variations in color-environment preferences within evidence-based frameworks.

4.3 Product & Packaging Design

Consumer color research (Labrecque & Milne, 2012; Singh, 2006; Aslam, 2006) validates systematic color-purchasing correlations across product categories and cultural contexts. The Color Experience System integrates this established consumer research through systematic cultural-product analysis enabling evidence-based packaging color selection.

Integration of documented color-appetite research (Spence et al., 2010) through semantic concept mapping (taste-dominant sensory identification) enables systematic application of validated food-color associations across cultural contexts.

4.4 Digital Interface Optimization

Interface color research (Bonnardel et al., 2011; Hill & Scharff, 1997; Coursaris et al., 2008) documents systematic color-usability correlations without providing integrated cultural adaptation methodology. The system enables systematic application of these validated findings through cultural cluster parameters addressing documented regional interface preferences.

5. Comparative Analysis

5.1 Color Experience System versus Existing Systems

Contemporary color classification systems demonstrate fundamental limitations in professional application contexts. Traditional approaches including CIELAB, Munsell, and NCS systems provide technical color specification without systematic integration of color-psychology research or cultural differentiation capabilities.

The Color Experience System addresses these limitations through systematic research integration enabling evidence-based color selection rather than purely technical color specification. While existing systems focus on color reproduction accuracy, this methodology emphasizes systematic application of documented color-psychology relationships in professional contexts.

5.2 Methodological Foundation

The methodology demonstrates systematic organization of 1,395 calculated color-semantic relationships across 60 operational concepts derived from established color research.

Complete matrix implementation encompasses systematic correlation between 15 color types, 12 sensory modalities, and 6 cultural clusters through empirically-validated confidence weighting.

This systematic foundation differentiates from intuitive color selection approaches through quantifiable parameters enabling reproducible professional color selection processes across diverse application contexts.

5.3 Integration Capabilities

The systematic methodology provides comprehensive integration capabilities addressing documented limitations in existing color systems. Cultural adaptation through validated cluster parameters, neuropsychological integration through multi-sensory frameworks, and professional application through context-specific analysis represent systematic advantages over traditional color classification approaches.

5.4 Comparative Framework Analysis

System	Cultural Adaptation	Neuropsychological Integration	Professional Application	Research Foundation
CIELAB	None	Visual only	Technical specification	Colorimetric standards

System	Cultural Adaptation	Neuropsychological Integration	Professional Application	Research Foundation
Munsell	Limited	Visual only	Art/Design focused	Perceptual uniformity
NCS	None	Visual only	Industrial specification	Natural color system
Pantone	Commercial	Visual only	Graphic design	Color reproduction
Color Experience System	6 validated clusters	12-sense integration	Multi-domain systematic	39 peer-reviewed sources

6. Theoretical Contributions

6.1 Neuropsychological Integration Framework

This methodology contributes to color science through systematic integration of neuropsychological research with practical color application. While existing approaches focus exclusively on visual perception parameters, this framework incorporates multi-sensory integration through Steiner's 12-sense systematic organization.

The integration of established cross-modal perception research (Ramachandran & Hubbard, 2001; Nakamura et al., 2008) through systematic sensory frameworks provides theoretical foundation for comprehensive color-experience correlation absent from traditional color classification systems.

6.2 Cultural Differentiation Theory

The research advances cultural color theory through empirically-validated culture cluster development. While cross-cultural color research documents regional variations (Berlin & Kay, 1969; Adams & Osgood, 1973), systematic cultural framework implementation enables practical application of these theoretical findings.

The systematic cultural framework translates documented cross-cultural color research into operational parameters, contributing to color theory through cultural-professional application integration rather than purely academic color variation documentation.

6.3 Theory-Practice Integration

This methodology contributes to color science through systematic bridging of theoretical color research with professional application requirements. The systematic integration enables evidence-based professional color selection processes replacing intuitive approaches with research-validated methodologies.

The framework demonstrates systematic organization of established color research enabling reproducible, culturally-appropriate color selection processes. This theoretical contribution addresses the documented gap between extensive color research and limited practical implementation frameworks in professional contexts.

7. Conclusion & Future Implications

7.1 Professional Color Science Applications

This methodology contributes to professional color science through systematic integration of established color research with practical application frameworks. The systematic approach enables evidence-based color selection processes across healthcare, architecture, packaging, and digital interface design contexts.

The research demonstrates systematic organization of validated color-psychology studies enabling reproducible professional color selection rather than intuitive approaches. This contribution addresses documented gaps between extensive color research and limited practical implementation methodologies in professional contexts.

7.2 Academic Research Directions

Future research priorities include expanded empirical validation of cultural cluster parameters across additional geographic regions and demographic populations. Systematic validation of neuropsychological color-sense correlations through controlled experimental frameworks represents priority research directions.

Integration with emerging color reproduction technologies and expanded validation across additional professional domains provide systematic research development opportunities. Cross-validation studies comparing systematic methodology outcomes with traditional color selection approaches offer empirical validation priorities.

7.3 Technological Integration Opportunities

The systematic framework enables integration with digital color management systems and automated design tools through systematic parameter specification. Cultural adaptation algorithms incorporating validated culture cluster parameters provide technological development opportunities for color-sensitive applications.

Machine learning integration through systematic color-context correlation training datasets represents future technological development directions. The methodology provides foundation for culturally-adaptive color systems addressing global design and branding requirements through evidence-based systematic approaches.

The systematic integration of established color research through practical application frameworks contributes to color science methodology while enabling professional implementation of validated color-psychology findings.

8. References

Historical Foundation & Color Theory Classics

- Albers, J. (1963). *Interaction of Color*. Yale University Press.
- Berlin, B., & Kay, P. (1969). *Basic Color Terms: Their Universality and Evolution*. University of California Press.
- Goethe, J.W. (1840). *Theory of Colors*. London: John Murray.
- Itten, J. (1961). *The Art of Color*. New York: Reinhold Publishing.
- Munsell, A.H. (1905). *A Color Notation*. Boston: Ellis.

Technical Standards & Colorimetry

- ASTM D7195-21. (2021). *Standard Guide for Setting Object Color Specifications*. ASTM International.
- ASTM E308-18. (2018). *Standard Practice for Computing the Colors of Objects by Using the CIE System*. ASTM International.
- ASTM E2022-16. (2016). *Standard Practice for Calculation of Weighting Factors for Tristimulus Integration*. ASTM International.
- CIE 15:2018. (2018). *Colorimetry, 4th Edition*. Commission Internationale de l'Éclairage.
- ISO 11664-4:2019. (2019). *Colorimetry — Part 4: CIE 1976 Lab Colour space**. International Organization for Standardization.
- Color Perception & Psychology Research
- Adams, F.M., & Osgood, C.E. (1973). A cross-cultural study of the affective meanings of color. *Journal of Cross-Cultural Psychology*, 4(2), 135-156.
- Aslam, M.M. (2006). Are you selling the right colour? A cross-cultural review of colour as a marketing cue. *Journal of Marketing Communications*, 12(1), 15-30.
- Bonnardel, N., Piolat, A., & Le Bigot, L. (2011). The impact of colour on website appeal and users' cognitive processes. *Displays*, 32(2), 69-70.
- Boynton, R.M., & Wagner, H.G. (1972). Color differences assessed by the minimally-distinct border method. *AIC 1972 Proceedings*, Stockholm.
- Griffin, L.D. (2005). Optimality of the basic colour categories for classification. *Journal of the Royal Society Interface*, 3(6), 71-85.

Hill, A., & Scharff, L.V. (1997). Readability of websites with various foreground/background color combinations, font types and word styles. *Proceedings of the 11th National Convention*, 742-746.

Cross-Cultural & Neuropsychological Studies

Goodwin, C. (2001). Practices of color classification. *Cognition and Communication at Work*, 17-40.

Coursaris, C.K., Swierenga, S.J., & Watrall, E. (2008). An empirical investigation of color temperature and gender effects on web aesthetics. *Journal of Usability Studies*, 3(3), 103-117.

Küller, R., Mikellides, B., & Janssens, J. (2009). Color, arousal, and performance—A comparison of three experiments. *Color Research & Application*, 34(2), 141-152.

Liu, Y., Cui, G., Luo, M.R., & Rigg, B. (2001). A comparative study of the perceived lightness for related and unrelated colours. *Color Research & Application*, 26(5), 365-374.

Nakamura, K., Kawabata, H., Osaka, N., & Osaka, M. (2008). The role of the right prefrontal cortex in recognition of facial expressions: A near-infrared spectroscopy study. *Brain and Cognition*, 66(1), 44-52.

Ou, L.C., & Luo, M.R. (2006). A colour harmony model for two-colour combinations. *Color Research & Application*, 31(3), 191-204.

AIC Proceedings - Color Science Research

Ishida, T., & Tanaka, H. (2008). Simulation of emerging color arrangement of cityscape and psychological evaluation. *AIC 2008 Proceedings*, Stockholm.

McLaren, K. (1972). Multiple linear regression: A new technique for improving colour difference formulae. *AIC 1972 Proceedings*, Stockholm.

Nayatani, Y. (2005). Brightness-to-lightness conversion and the Helmholtz-Kohlrausch effect. *Color Research & Application*, 30(5), 330-343.

Richter, K. (1972). Description of colour attributes and colour differences. *AIC 1972 Proceedings*, Stockholm.

Um, J., Eum, K., & Lee, J. (2002). A study of the emotional evaluation models of color patterns based on the adaptive fuzzy system and the neural network. *Color Research & Application*, 27(4), 267-278.

Vos, J.J., & Walraven, P.L. (1972). Zone-fluctuation line element describing colour discrimination. *AIC 1972 Proceedings*, Stockholm.

Applied Color Research

Birren, F. (1988). *Light, Color and Environment*. New York: Van Nostrand Reinhold.

Dijkstra, K., Pieterse, M.E., & Pruyn, A. (2006). Physical environmental stimuli that turn healthcare facilities into healing environments through psychologically mediated effects. *Journal of Environmental Psychology*, 26(2), 166-181.

Labrecque, L.I., & Milne, G.R. (2012). Exciting red and competent blue: the importance of color in marketing. *Journal of the Academy of Marketing Science*, 40(5), 711-727.

Mahnke, F.H. (1996). *Color, Environment, and Human Response*. New York: John Wiley & Sons.

Palmer, S.E., & Schloss, K.B. (2010). An ecological valence theory of human color preference. *Proceedings of the National Academy of Sciences*, 107(19), 8877-8882.

Ramanath, R., Kuehni, R.G., Snyder, W.E., & Hinks, D. (2004). Spectral spaces and color spaces. *Color Research & Application*, 29(1), 29-37.

Singh, S. (2006). Impact of color on marketing. *Management Decision*, 44(6), 783-789.

Spence, C., Levitan, C.A., Shankar, M.U., & Zampini, M. (2010). Does food color influence taste and flavor perception in humans? *Chemosensory Perception*, 3(1), 68-84.

Stone, N.J. (2003). Environmental view and color for a simulated telemarketing task. *Journal of Environmental Psychology*, 23(1), 63-78.

Valdez, P., & Mehta, A. (1994). Effects of color on emotions. *Journal of Experimental Psychology: General*, 123(4), 394-409.

Zhang, H., & Montag, E.D. (2006). How well can people use different color attributes? *Color Research & Application*, 31(6), 445-457.

Neuropsychological & Synesthesia Research

Ramachandran, V.S., & Hubbard, E.M. (2001). Synesthesia—a window into perception, thought and language. *Journal of Consciousness Studies*, 8(12), 3-34.

About the Color Experience System

The Color Experience System was developed through systematic analysis of contemporary color science research, neuropsychological studies, and cross-cultural color perception data.

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Document Classification: Academic Publication - Open Access

First Publication: august 8, 2025 - xconea.com

Version 1.0 - Intellectual Property Documentation

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